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our win

Ex# 4-6

(51)

① Find G.M b/w

(iv) $-2i$ and $8i$

(i) -2 and -8

Sol

$$\therefore G = \pm \sqrt{ab}$$

here, $a = -2$ & $b = -8$

$$\therefore G = \pm \sqrt{(-2)(-8)}$$

$\Rightarrow G = \pm \sqrt{16} = \pm 4$ is the
required G.M b/w -2 & -8 .

Sol

$$\therefore G = \pm \sqrt{ab}$$

here, $a = -2i$, $b = 8i$

$$\therefore G = \pm \sqrt{(-2i)(8i)}$$

$$\Rightarrow G = \pm \sqrt{-16i^2} = \pm \sqrt{16}$$

$\Rightarrow G = \pm 4$ is the
Required G.M.

(ii) 3 and $1/3$

Sol

$$\therefore G = \pm \sqrt{ab}$$

here, $a = 3$, $b = 1/3$

$$\therefore G = \pm \sqrt{(3)(1/3)}$$

$$\Rightarrow G = \pm \sqrt{1} = \pm 1$$

It is the required G.M
b/w 3 and $1/3$.

(v) $7+i$ and $7-i$

Sol

$$\therefore G = \pm \sqrt{ab}$$

here, $a = 7+i$, $b = 7-i$

$$\therefore G = \pm \sqrt{(7+i)(7-i)}$$

$$\Rightarrow G = \pm \sqrt{(7)^2 - (i)^2} = \pm \sqrt{49 - i^2}$$

$$\Rightarrow G = \pm \sqrt{49+1} = \pm \sqrt{50}$$

$\Rightarrow G = \pm 5\sqrt{2}$ is the
required G.M.

(iii) $8\sqrt{2}$ and $9\sqrt{2}$

Sol

$$\therefore G = \pm \sqrt{ab}$$

here, $a = 8\sqrt{2}$, $b = 9\sqrt{2}$

$$\therefore G = \pm \sqrt{(8\sqrt{2})(9\sqrt{2})}$$

$$\Rightarrow G = \pm \sqrt{72(2)}$$

$$\Rightarrow G = \pm \sqrt{144} = \pm 12$$

(vi) $16/9$ and $9/25$

Sol

$$\therefore G = \pm \sqrt{ab}$$

here, $a = 16/9$, $b = 9/25$

$$\therefore G = \pm \sqrt{(16/9)(9/25)}$$

$$\Rightarrow G = \pm \sqrt{16/25}$$

$$\Rightarrow G = \pm 4/5$$

Insert:

(i) Two G.Ms b/w 3 and 81

S.1

let G_1 and G_2 be two G.Ms
b/w 3 and 81.

$$\therefore a_n = ar^{n-1}$$

here, $a = 3$, $n = 2$, $a_n = 81$.

$$\therefore 81 = (3)(r)^{2-1}$$

$$\Rightarrow 27 = r^3$$

$$\Rightarrow r = 3$$

Now,

$$G_1 = ar = (3)(3) = 9$$

$$G_2 = ar^2 = (3)(3)^2 = 27$$

(ii) Three G.Ms b/w 2 and $\frac{1}{2}$

S.1

let G_1, G_2 and G_3 be 3
G.Ms b/w 2 and $\frac{1}{2}$

$$\therefore a_n = ar^{n-1}$$

here, $a = 2$, $a_n = \frac{1}{2}$, $n = 5$

$$\therefore \frac{1}{2} = (2)(r)^{5-1}$$

$$\Rightarrow r^4 = \frac{1}{4}$$

$$\Rightarrow r = \left(\frac{1}{4}\right)^{\frac{1}{4}}$$

$$\Rightarrow r = \frac{1}{\sqrt{2}}$$

(S2)

$$G_1 = ar = (2)\left(\frac{1}{\sqrt{2}}\right) = \frac{2}{\sqrt{2}}$$

$$G_2 = ar^2 = (2)\left(\frac{1}{\sqrt{2}}\right)^2 = 1$$

$$G_3 = ar^3 = (2)\left(\frac{1}{\sqrt{2}}\right)^3 = \frac{1}{\sqrt{2}}$$

(iii) Four G.Ms b/w 3 and 96

S.1

let G_1, G_2, G_3 and G_4 be 4
G.Ms b/w 3 and 96.

$$= a_n = ar^{n-1}$$

here, $a = 3$, $a_n = 96$, $n = 6$

$$\therefore 96 = (3)(r)^{6-1}$$

$$\Rightarrow 32 = r^5$$

$$\Rightarrow r = 2$$

Now,

$$G_1 = ar = (3)(2) = 6$$

$$G_2 = ar^2 = (3)(2)^2 = 12$$

$$G_3 = ar^3 = (3)(2)^3 = 24$$

$$G_4 = ar^4 = (3)(2)^4 = 48$$

(iv) Five G.Ms b/w 16 and 1024

S.1

let G_1, G_2, G_3, G_4 and G_5 be
5 G.Ms b/w 16 and 1024

$$\therefore a_n = ar^{n-1}$$

here, $a = 16$, $n = 7$, $a_n = 1024$

$$\therefore 1024 = (16)(r)^{7-1}$$

(53)

$$\Rightarrow 64 = 8^2$$

$$\Rightarrow r = 2$$

Now,

$$G_1 = ar = (16)(2) = 32$$

$$G_2 = ar^2 = (16)(2)^2 = 64$$

$$G_3 = ar^3 = (16)(2)^3 = 128$$

$$G_4 = ar^4 = (16)(2)^4 = 256$$

$$G_5 = ar^5 = (16)(2)^5 = 512$$

There are the req. 5
G.Ms b/w 16 and 1024. sl.

③ A.M b/w 2 no. is 29 & G.M is

21 - Find the no.s.

sl.

let 2 no. be a and b.

A/c to given condition:

$$\Rightarrow A.M = 29 \text{ or}$$

$$\Rightarrow \frac{a+b}{2} = 29 \text{ --- (1)}$$

$$\Rightarrow G.M = 21 \text{ or}$$

$$\Rightarrow \sqrt{ab} = 21$$

$$\Rightarrow ab = 441 \text{ --- (2)}$$

Take (1).

$$\Rightarrow a+b = 58 \text{ (2)}$$

$$\Rightarrow a+b = 58$$

$$\Rightarrow a = 58 - b \text{ --- (3)}$$

Put a in (2)

$$\therefore (2) \Rightarrow (58-b)b = 441$$

$$\Rightarrow -b^2 + 58b = 441$$

$$\Rightarrow b^2 - 58b + 441 = 0$$

$$\Rightarrow b^2 - 49b - 9b + 441 = 0$$

$$\Rightarrow b(b-49) - 9(b-49) = 0$$

$$\Rightarrow (b-49)(b-9) = 0$$

either,

$$b = 49 \text{ or } b = 9$$

Put b in (3)

$$\text{At } b = 49$$

$$\Rightarrow a = 58 - 49$$

$$\Rightarrow a = 9$$

$$\text{At } b = 9$$

$$\Rightarrow a = 58 - 9$$

$$\Rightarrow a = 49$$

④ Find n when $\frac{a^{n-2} + b^{n-2}}{a^{n-3} + b^{n-3}}$ is a G.M
b/w a and b.

sl.

A/c to given condition:

$$(ab)^{\frac{n-2}{2}} = \frac{a^{n-2} + b^{n-2}}{a^{n-3} + b^{n-3}}$$

(54)

$$\Rightarrow a^{\frac{1}{2}} b^{\frac{1}{2}} (a^{n-3} + b^{n-3}) = a^{n-2} + b^{n-2}$$

$$\Rightarrow a^{n-\frac{5}{2}} b^{\frac{1}{2}} + b^{n-\frac{5}{2}} a^{\frac{1}{2}} = a^{n-2} + b^{n-2}$$

$$\Rightarrow b^{n-\frac{5}{2}} a^{\frac{1}{2}} - b^{n-2} = a^{n-2} - a^{n-\frac{5}{2}} b^{\frac{1}{2}}$$

$$\Rightarrow b^{n-\frac{5}{2}} (a^{\frac{1}{2}} - b^{\frac{1}{2}}) = a^{n-2} (a^{\frac{1}{2}} - b^{\frac{1}{2}})$$

$$\Rightarrow 1 = \frac{a^{n-\frac{5}{2}}}{b^{n-\frac{5}{2}}}$$

$$\Rightarrow \left(\frac{a}{b}\right)^{n-\frac{5}{2}} = \left(\frac{a}{b}\right)^0$$

Thus,

$$\Rightarrow n - \frac{5}{2} = 0$$

$$\Rightarrow n = \frac{5}{2} \text{ A.P.}$$

(6) The A.M of 2 +ve integral no.s exceed their +ve G.M by 2 & their sum is 20. Find the no.s. s.t

Ans let 2 no.s be a & b

Atc to given condition:

$$\Rightarrow \text{A.M} = \text{G.M} + 2 \text{ or } \dots$$

$$\Rightarrow \frac{a+b}{2} = \sqrt{ab} + 2 \rightarrow \text{①}$$

$$\therefore a+b = 20$$

$$\Rightarrow a = 20 - b \rightarrow \text{②}$$

$$\text{①} \Rightarrow \frac{20}{2} = \sqrt{ab} + 2$$

$$\Rightarrow 10 = \sqrt{ab} + 2$$

$$\Rightarrow \sqrt{ab} = 8$$

$$\Rightarrow ab = 64 \rightarrow \text{③}$$

Put a in above eq.

$$\Rightarrow b(20-b) = 64$$

$$\Rightarrow 20b - b^2 = 64$$

$$\Rightarrow b^2 - 20b + 64 = 0$$

$$\Rightarrow b^2 - 16b - 4b + 64 = 0$$

$$\Rightarrow b(b-16) - 4(b-16) = 0$$

$$(b-16)(b-4) = 0$$

either,

$$b = 16 \text{ or } b = 4$$

Put b in ②

$$\text{At } b = 16$$

$$\Rightarrow a = 20 - 16 = 4$$

$$\text{At } b = 4$$

$$\Rightarrow a = 20 - 4 = 16$$

Thus req 2 no.s are 4 and 16. A.P.

(55)

(7) The A.M b/w 2 no.s is 5 and their true G.M is 4. Find the numbers.

Sol

let a and b be 2 nos.

A/c to given condition:

$$\Rightarrow A.M = 5 \text{ or}$$

$$\Rightarrow \frac{a+b}{2} = 5$$

$$\Rightarrow a+b = 10$$

$$\Rightarrow a = 10 - b \text{ — (1)}$$

$$\Rightarrow G.M = 4$$

$$\Rightarrow \sqrt{ab} = 4$$

$$\Rightarrow ab = 16 \text{ — (2)}$$

Put a in (2)

$$\Rightarrow b(10-b) = 16$$

$$\Rightarrow 10b - b^2 = 16$$

$$\Rightarrow b^2 - 10b + 16 = 0$$

$$\Rightarrow b^2 - 8b - 2b + 16 = 0$$

$$\Rightarrow b(b-8) - 2(b-8) = 0$$

$$\Rightarrow (b-8)(b-2) = 0$$

either,

$$b = 8 \text{ or } b = 2$$

put b in (1)

$$\text{At } b = 8$$

$$\Rightarrow a = 10 - 8 = 2$$

$$\text{At } b = 2$$

$$\Rightarrow a = 10 - 2 = 8$$

Ans.